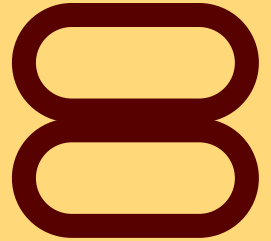
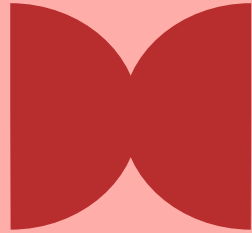


CTS2000: Scripting for the Humanities



Merve Tekgürler
September 14, 2026

/Meyr-Veh/

How to pronounce my name



Stanford University – History Corner, ca. 1996

Welcome to CTS2000!

This course introduces students to core concepts in data creation, representation, and interpretation. In addition to thinking critically about the data as a form of representation, students will also analyse the ethical considerations of data and develop a number of models for the ethical contextualization of data. Students will learn to model data in software, understand the history of the concept of data, develop a data schema, and critique data as representation. Students will develop basic programming skills in the Python programming language. They will create data structures, use programming libraries to manipulate and work with data, and develop their own functions.



Learning Goals of this Course



Python

We will learn how to code.



Digital and Computational Humanities

We will learn about digital and computational research in the humanities.



Project-Driven Research

We will learn how to develop a research project, with an emphasis on critical data analysis.

Assignments

1

Participation

Participation in the class combines both being an active participant during the class meetings and preparing before/after the course.

2

Individual Project

The purpose of the individual project is to build necessary analytical skills to study the code and digital approaches behind various projects.

3

Group Project

The group project will require you to work in small groups (3-4 students) and develop a project that answers a clearly defined question using computational methods.

Assignments and Grading

- Participation 20%
- Individual Project
 - Short Paper (Literature Review) 20%; due October 30
 - Presentation (Code Reproduction and Peer Review) 10%; due November 4 and 6
- Group Project
 - Topic Statement (Scope, Questions, Dataset) 10%; due October 9
 - Long Paper (Research Findings) 20%; due December 7
 - Repository (Original Code and Annotation) 10%; due December 7
 - Presentation (In-Class Conference style) 10%; due last week of class

Participation

Students will take part in class discussions both during class lectures and labs as well as on the class Slack workplace. **Adequate participation requires completing all assigned readings before the start of that day's class meeting and practicing coding outside of class hours.** Coding practice includes repeating the coding exercises completed in the class and working through the unfinished sections of the notebooks following the session in which they were introduced. There will be a Slack channel for each week, where students are expected to reflect on their coding practice, ask each other questions and help each other find solutions.

Regular participation in the group project, both on Slack and in person, also counts toward the participation grade (see Group Project Check-in Weeks 8–11). These check-ins can be short updates on your group's Slack channel or project repository. We will discuss best practices for check-ins once the project groups are formed.

Individual Project

To build a strong foundation in digital humanities, you need to become familiar with how digital humanists use Python in their research. This involves studying and analyzing code written by other researchers, as well as reproducing their workflows to understand both the technical and interpretive dimensions of their work. It also means critically evaluating the projects you study—examining their data sources, methodological approaches, design choices, and coding decisions—to assess their scholarly rigor, transparency, and impact.

Individual Project

Short Paper (20%)

You will choose a topic, either a methodology such as topic modeling or a task such as named entity recognition, and identify three papers on that topic. You will briefly summarize their methods, and the reasons the authors chose them, and study their code and data. You will then critically evaluate the papers to identify their weaknesses and strengths. Your short paper should be 3 pages long.

Individual Project

Presentation (10%)

You will replicate the code from one of the papers you evaluated and walk us through it. To do this, you will create a Jupyter Notebook and get the research code running in our course environment. You will record yourself running the code and explaining each step to an audience unfamiliar with your topic but familiar with everything that we have covered in this class. Your presentations should be 5–6 mins long. You will upload your presentation and your Jupyter Notebook to CourseLink.

Once you upload your presentation, you will be matched with two other students. You are expected to watch their presentations carefully and ask constructive questions. We will discuss how to give feedback and clarify the details about the matching process later in the semester. The quality of your engagement with your colleagues' work is an essential component of this assignment.

Group Project

Completing a project from start to finish is an important learning goal of this course. You are expected to design a project that asks a concrete research question, situates that question within the secondary scholarship, and identifies a dataset that will let you answer the question with computational methods. Keeping in mind that this is an introductory course, you should focus on one question that interests you and that will allow you to meet the learning objectives of the course. **You are not expected to develop a new theory or to cover everything on a given topic or dataset. The success of the assignment depends on whether the dataset and the computational methods you have chosen actually answer the research question you put forward.**

Collaboration is an essential part of digital humanities. Another goal of this assignment is to teach you how to work in small groups of 3–4 students. You will be graded as a group but you will also be asked to clarify your individual contributions to the project in various ways.

Group Project

Topic Statement (10%)

Topic statement is the first assignment for the group project. It should describe the motivating research idea, include your initial research question(s) and identify the dataset. Please offer a well-grounded reasoning on why this dataset is relevant for this research direction. We will discuss how to evaluate datasets in class. You must identify all the members of your group beforehand and devise a plan for the division of labor (not included in the page count). Division of labor should not mean one student doing the writing and another doing the coding. You must split the tasks equitably. Upon forming your groups, you will be put into a Slack channel with your group members. The topic statement should be 2 pages long.

Group Project

Long Paper (20%)

You will write a long paper (5–7 pages) summarizing your findings. This paper should clearly identify the question you studied and your approach to it. You must describe the data and the methods you used to answer this question. What is your answer to the question? How do you support it with data? Your paper must include at least one data visualization and explanation of its significance. Address the limitations of your approach. You should highlight where your methods were not successful or where the data was lacking. What are some potential future research directions?

This is a collaborative paper, which means your group must develop a shared voice. While you may divide the writing of individual sections among team members, everyone is collectively responsible for the quality and coherence of the final product.

The paper should include a bibliography and an author contribution statement in which each student writes a sentence or two describing their contributions to the paper and the project as a whole (not included in the page count).

Group Project

Repository (10%)

Coding for the group projects should be done on GitHub. The idea behind the repository is to give you the opportunity to test different approaches and experiment with new directions in your research. You are expected to use this repository regularly and report on intermediate results in your group's Slack channel. GitHub is the industry standard for collaborative coding and learning how to use it effectively is an integral part of this assignment. It also allows you to show your individual contributions to the group project, which will be taken into consideration during the grading.

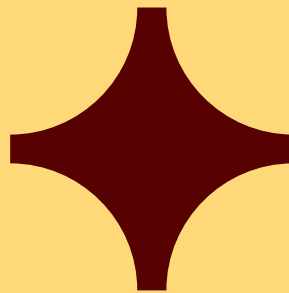
Group Project

Presentation (10%)

Your group will give a 10-minute presentation of your project using a Jupyter Notebook. You will create one notebook from your code base and run it live during the presentation. Your notebook should have three sections. The first section will introduce your dataset and your question. The second will contain your methods. The third will answer your question supported by data analysis and visualization.

The presentations will be held during the last week of the course in a mini conference format. Each group will be assigned a 20-minute presentation slot, 10 minutes for their presentation and 10 minutes for the Q&A. Each group will also be matched with another group presenting on a different day and will be responsible for kickstarting the Q&A after that group's presentation. Groups will be evaluated on the quality of their engagement with the work of their assigned group.

Any Questions?



Introductions

**Find someone in
the class that you
never met before
and introduce
yourself!**

What is your name and what are your pronouns?

What are you majoring in and what year are you?

What is your experience with coding?

What is your experience with (digital) humanities and CTS?

What are you curious or excited about in this class?



Artificial Intelligence

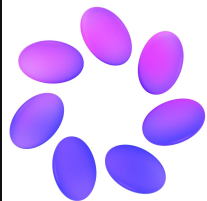
What is generative AI?



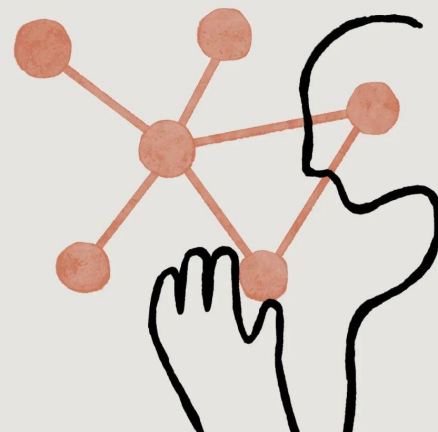
Gemini



cohere



Meta AI



ANTHROPIC

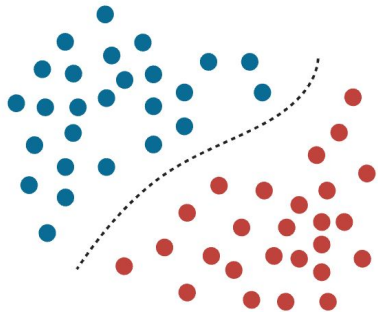
Let's take a step back: What is AI?

Discriminative vs Generative AI Models

Discriminative (classic)

Predict a label/class given the features of input data

What's learned?: Decision boundary

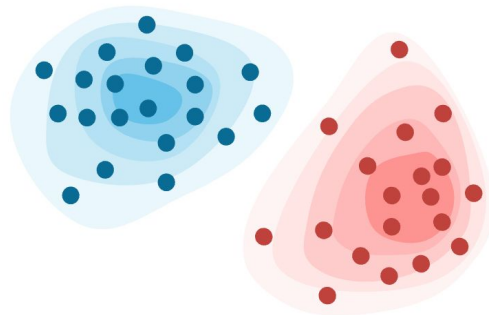


Determine if this email is spam or not spam

Generative

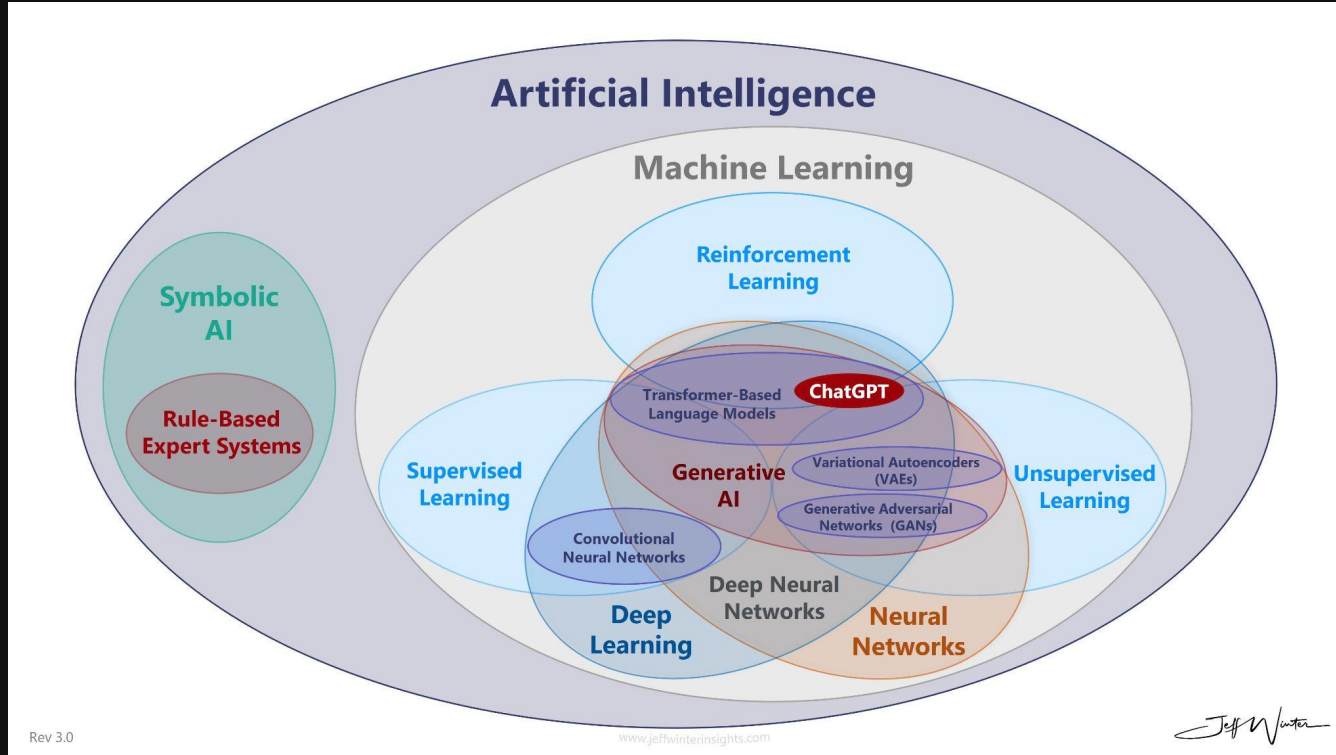
Abstract underlying patterns in input data in order to generate new content

What's learned?: Probability distributions of the data

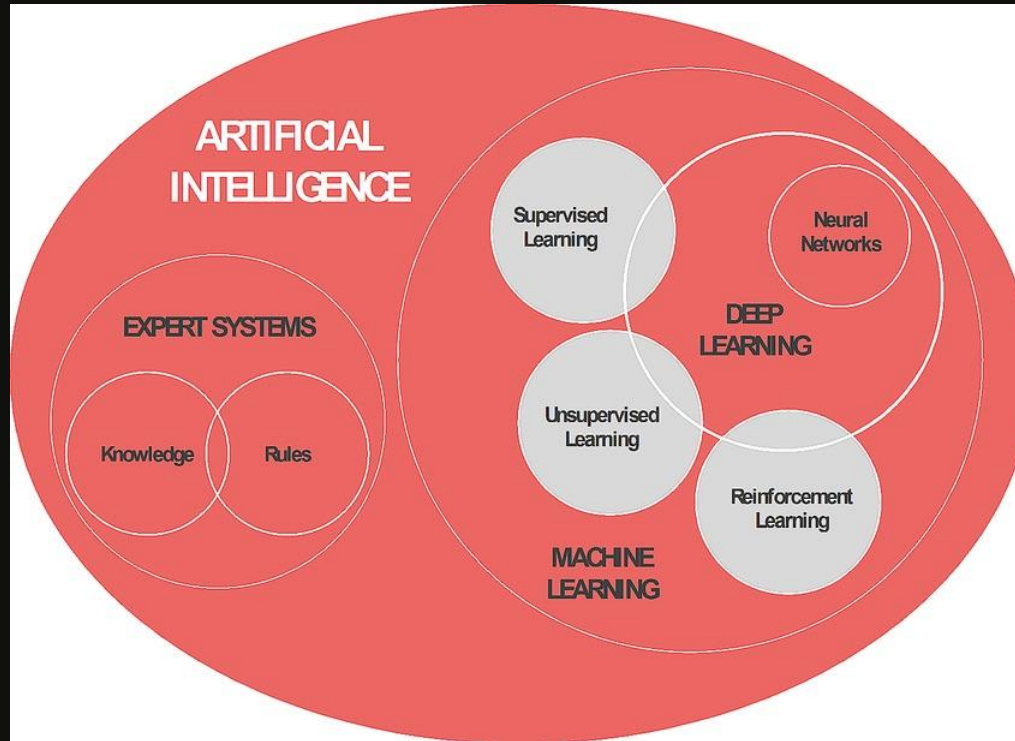


Create a spam email

Let's take a step back: What is AI?

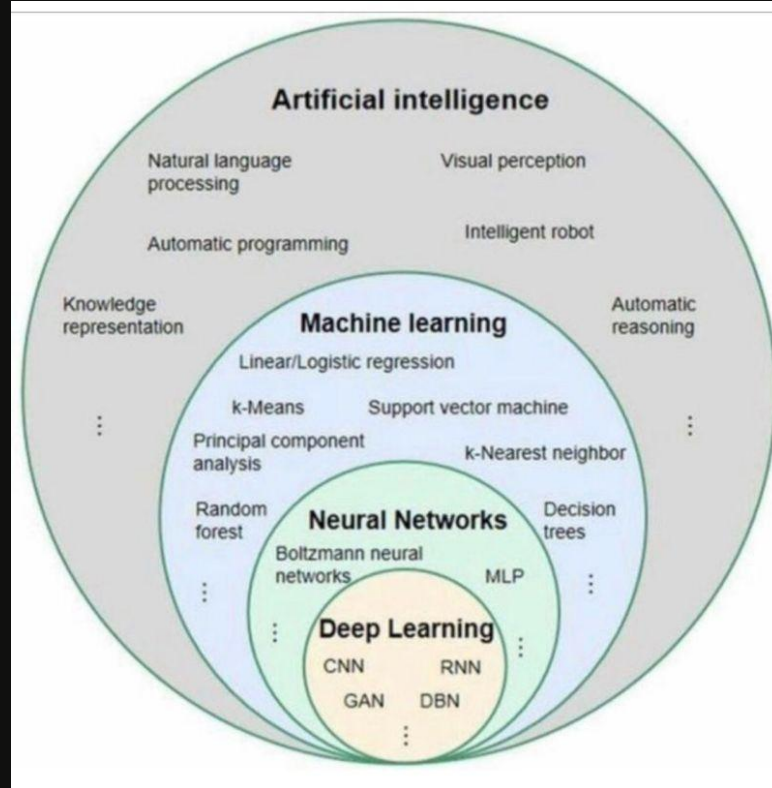


Let's take a step back: What is AI?

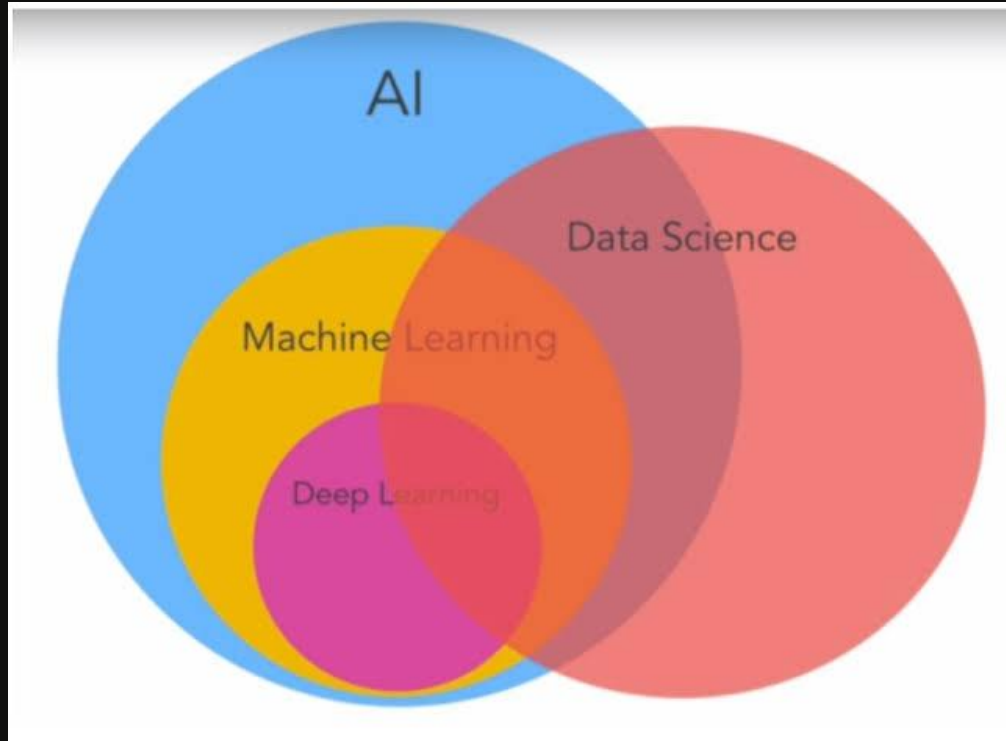


Let's take a step back: What is AI?

"Data Science: Defined By Venn Diagram"

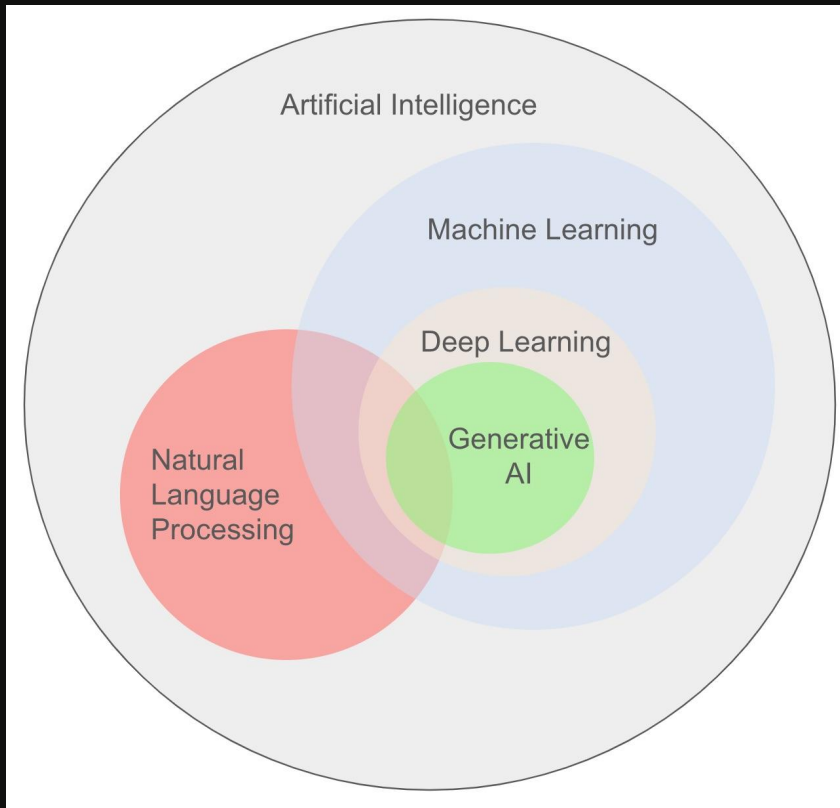


Let's take a step back: What is AI?



Let's take a step back: What is AI?

Natural Language
Processing/Computational
Linguistics



What is AI?: My definition

2017, Transformer paper

Attention Is All You Need

Ashish Vaswani*
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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.0 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature.

What is AI?: My definition

2020, GPT-3 paper

Language Models are Few-Shot Learners

Tom B. Brown*

Benjamin Mann*

Nick Ryder*

Melanie Subbiah*

OpenAI

Abstract

Recent work has demonstrated substantial gains on many NLP tasks and benchmarks by pre-training on a large corpus of text followed by fine-tuning on a specific task. While typically task-agnostic in architecture, this method still requires task-specific fine-tuning datasets of thousands or tens of thousands of examples. By contrast, humans can generally perform a new language task from only a few examples or from simple instructions – something which current NLP systems still largely struggle to do. Here we show that scaling up language models greatly improves task-agnostic, few-shot performance, sometimes even reaching competitiveness with prior state-of-the-art fine-tuning approaches. Specifically, we train GPT-3, an autoregressive language model with 175 billion parameters, 10x more than any previous non-sparse language model, and test its performance in the few-shot setting. For all tasks, GPT-3 is applied without any gradient updates or fine-tuning, with tasks and few-shot demonstrations specified purely via text interaction with the model. GPT-3 achieves strong performance on many NLP datasets, including translation, question-answering, and cloze tasks, as well as several tasks that require on-the-fly reasoning or domain adaptation, such as unscrambling words, using a novel word in a sentence, or performing 3-digit arithmetic. At the same time, we also identify some datasets where GPT-3's few-shot learning still struggles, as well as some datasets where GPT-3 faces methodological issues related to training on large web corpora. Finally, we find that GPT-3 can generate samples of news articles which human evaluators have difficulty distinguishing from articles written by humans. We discuss broader societal impacts of this finding and of GPT-3 in general.

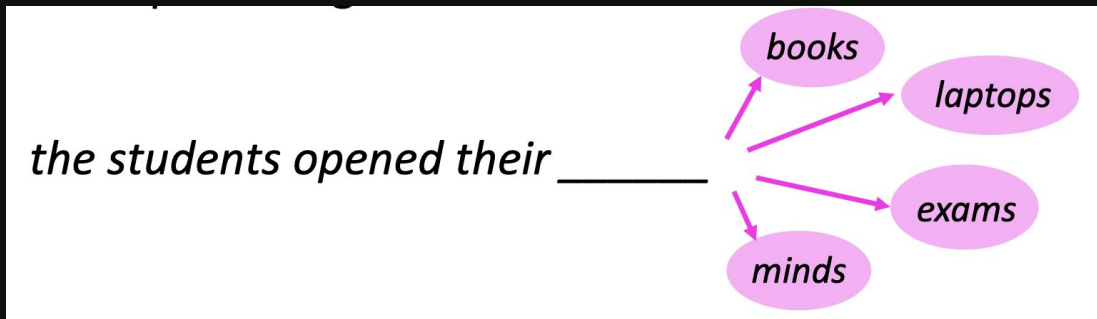
What is AI?: My definition

In the field of natural language processing (NLP), a language model “is a machine learning model that predicts upcoming words” based on the words that came before. Large language models (LLMs) are trained on enormous amounts of text, learning complicated relationships between words stored in billions of parameters. This increase in scale led to novel applications beyond word prediction, perhaps most significantly to the capacity for a computer to process instructions in human language. This is in and of itself a technological breakthrough. And somehow it was not enough. **LLMs became artificial intelligence (AI) and even artificial general intelligence.** The industry tells us that AI will cure cancer and all diseases so that we can live forever and also make human labor redundant and turn us all into paper clips.

I argue that AI is not a technology but a socially-constructed and politically-motivated frame for a collection of otherwise normal computer technologies. Separating the term ‘AI’ from LLMs and all the other technologies that are placed under its umbrella allows historians to sidestep having to justify or reject these technologies. (proposal submitted to AHR)

LLMs are autoregressive language models

Language Modeling is the task of predicting what word comes next



More formally: given a sequence of words compute the probability distribution of the next word, where the next word can be any word in the vocabulary of the model.

Statistics and Rule-Based Approaches

Above is a question from Dan Jurafsky's CS124 (Winter 2023) and below is an example of a context free grammar that I created for SYMSYS 200

We are given the following corpus, similar to the one in lecture but with "ham" replaced by "Sam" and "I am Sam" included twice:

- `<s> I am Sam </s>`
- `<s> Sam I am </s>`
- `<s> I am Sam </s>`
- `<s> I do not like green eggs and Sam </s>`

Using a bigram language model with add-one smoothing, what is $P(\text{Sam} \mid \text{am})$? Include `<s>` and `</s>` in your counts just like any other token.



History Discourse Bot
@HistBotSymsys

What does history mean?

5:00 AM · Apr 10, 2023 · 2 Views



```
MAIN -> SENTENCE "." | QUESTION "?"
-> " "
QUESTION -> QUESTIONWORD _ "is" _ OBJ | "How does" _ OBJ _ "relate to" _ OBJ
| "Why is" _ OBJ _ "relevant" | "How many" _ THINGS _ "are you" _ GERUND | "Why
is" _ OBJ _ "in your" _ THINGS | "What does" _ OBJ _ "mean" | QUESTIONWORD _
"is" _ OBJ _ ADJ
QUESTIONWORD -> "What" | "When" | "Where" | "Why" | "How"
SENTENCE -> SUB _ VERB _ MOD | FANCYPHRASES _ SUB _ VERB _ MOD
SUB -> "Scholars" | "Historians" | "Ranke" | "Foucault" | "Grafton" | "Darnton" | "Davis"
VERB -> "stated" | "discussed" | "evaluated" | "emphasized" | "showed" | "analyzed" |
"explained"
FANCYPHRASES -> "With that in mind," | "Based on this approach," | "Based on their"
_ THINGS | "In the 80s," | "At the turn of the century,"
GERUND -> "writing" | "presenting" | "reading and" _ GERUND | "reviewing"
MOD -> "historicity of" _ OBJ | "the significance of" _ OBJ | "the importance of" _ OBJ |
"the debates about" _ OBJ | OBJ _ "and" _ SENTENCE | OBJ _ "or" _ SENTENCE
ADJ -> "socially constructed" | "historically embedded" | "structural" | "embodied"
OBJ -> "discourse" | "representation" | "history" | "the past" | "historiography" | "an
historian" | "the field" | "their most recent work" | "capitalism" | "gender" | "a primary
source" | "humanities" | "secondary literature" | "practice" | "an archive" | "imagined
communities" | "content" | "cultural capital"
THINGS -> "monographs" | "articles" | "talks" | "edited volumes" | "papers"
```

Randomness is important

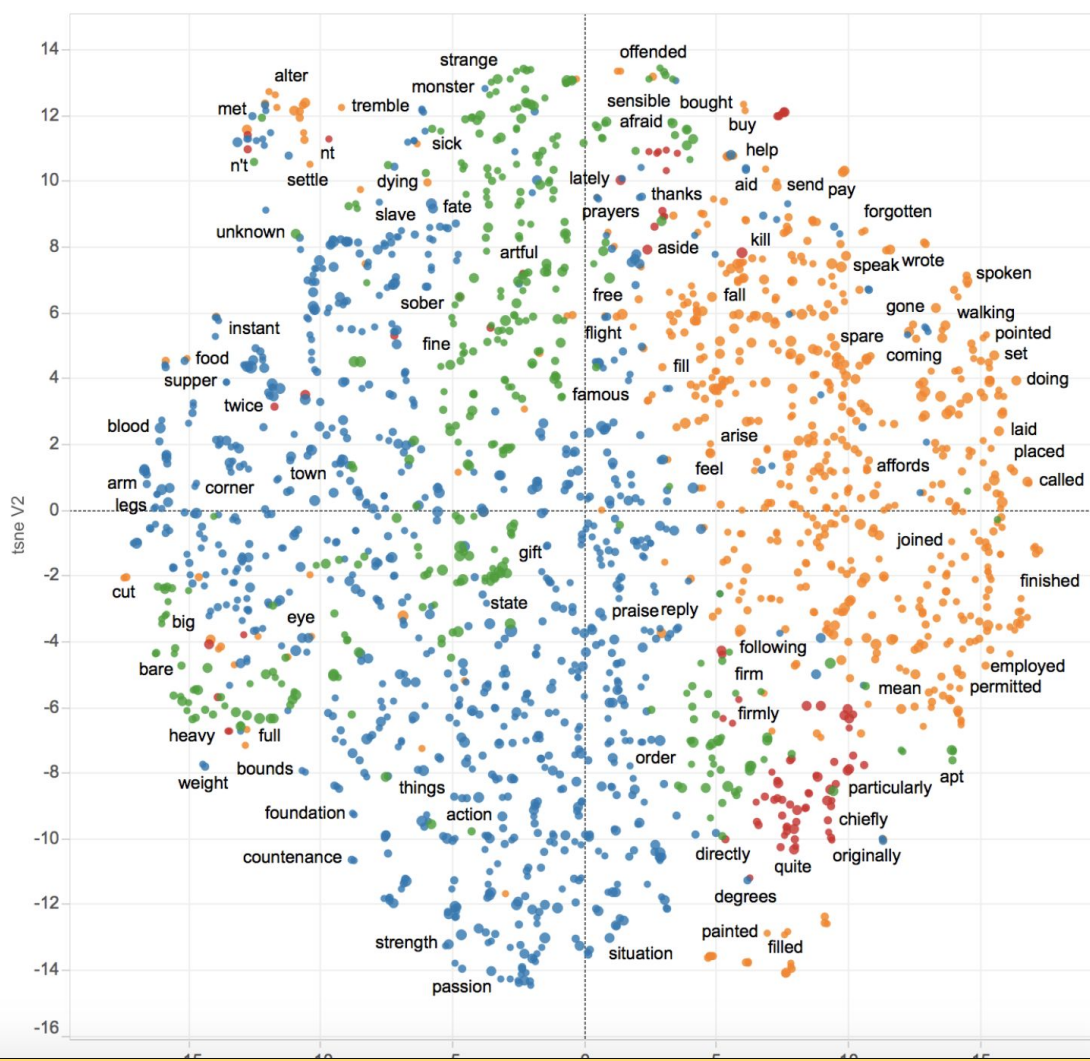
The best thing about AI is its ability to

learn	4.5%
predict	3.5%
make	3.2%
understand	3.1%
do	2.9%

Temperature is the parameter that determines how likely the model is to sample from less-probable options.

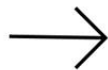
Word beddings

Word embeddings are numerical representations of words in a multi-dimensional space, enabling the capture of semantic relationships and similarities between them.



How do they learn to predict the next token?

But I do
think it is
their husbands'
faults if
wives do fall.



But I do
[dance] it is
their [MASK]
faults if
[MASK] do fall.

GPT-3 makeup (2020)

- 60% Common Crawl* (internet data) - 0.7x weight
- 22% WebText2 - 5x weight
- 8% Books1 - 1.2x weight
- 8% Books2 - 1.2x weight
- 3% Wikipedia (in English?) - 5x weight

* As of March 2023, in the most recent version of the Common Crawl dataset, 46% of documents had English as their primary language (followed by Russian, German, Chinese, Japanese, French and Spanish, all below 6%)

Fan Fiction and Social Media as Text Corpora

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The Girl in the Woods by [Kristen Ashley](#)
Price: \$5.99 USD. Words: 105,330. Language: English. Published: March 21, 2023.
Categories: [Fiction](#) » [Romance](#) » [Suspense, Fiction](#) » [Thriller & suspense](#)
» [Crime thriller](#)
His years tracking the Crystal Killer have taken their toll on Zachariah Lazarus. The latest victim was found in the sleepy town of Misted Pines. But something is not right. The victim's employer, Lucinda Bonner, decides she's going to do everything she can to help Rus navigate the intricate, and sometimes sordid, history of the town of Misted Pines.



Echo by [Sybil Bartel](#)
Series: [The Alpha Elite Series](#), Book 6. Price: \$5.99 USD. Words: 92,600.
Language: English. Published: March 28, 2023 by [Sybil Bartel](#). Categories: [Fiction](#) » [Romance](#) » [Contemporary, Fiction](#) » [Romance](#) » [Suspense, Fiction](#) » [Romance](#)
» [Military](#)
Navv SEAL. Mercenary. Ghost. Joining the military wasn't a choice. it was survival.



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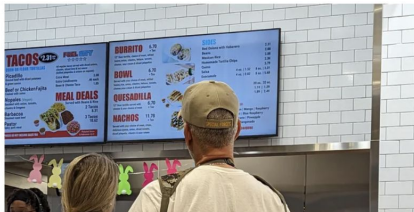
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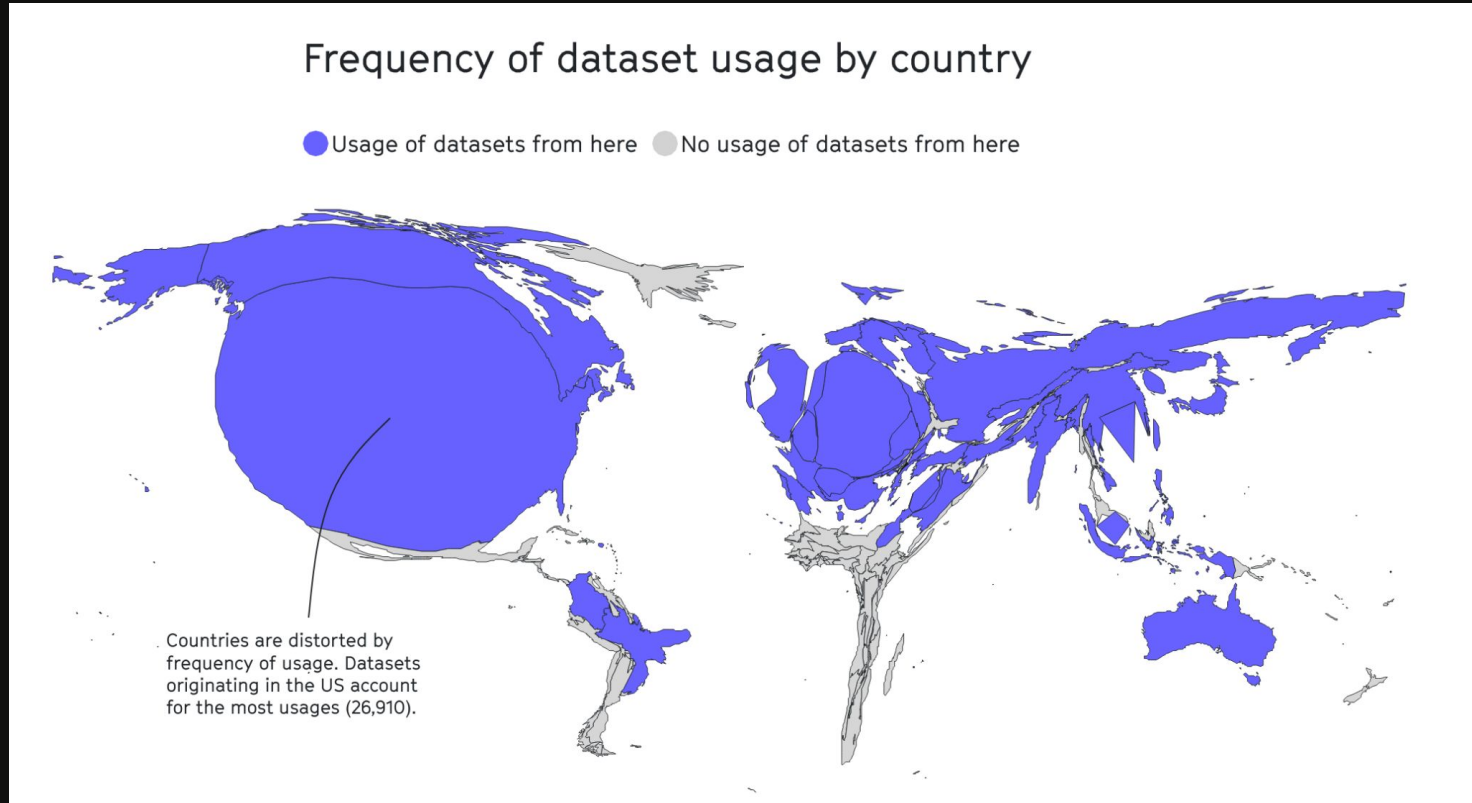
  [r/pics](#) · Posted by [u/SeventyFix](#) 18 hours ago    2 & 2 More [Join](#)

60.2k

Texas Open Carry Law: Having the barrel of a gun pointing at you while eating dinner



Frequency of data set usage based on point of origin (2022)



What are some of the ethical concerns with LLMs?

Timnit Gebru (DAIR), Emily M. Bender (University of Washington), Angelina McMillan-Major (University of Washington), Margaret Mitchell (Hugging Face)

March 31, 2023|

TL;dr: The harms from so-called AI are real and present and follow from the acts of people and corporations deploying automated systems. Regulatory efforts should focus on transparency, accountability and preventing exploitative labor practices.

Labor Practices

BUSINESS • TECHNOLOGY

Exclusive: OpenAI Used Kenyan Workers on Less Than \$2 Per Hour to Make ChatGPT Less Toxic



This image was generated by OpenAI's image-generation software, Dall-E 2. The prompt was: "A seemingly endless view of African workers at desks in front of computer screens in a printmaking style." TIME does not typically use AI-generated art to illustrate its stories, but chose to in this instance in order to draw attention to the power of OpenAI's technology and shed light on the labor that makes it possible. Image generated by Dall-E 2/OpenAI

Copyright

The New York Times

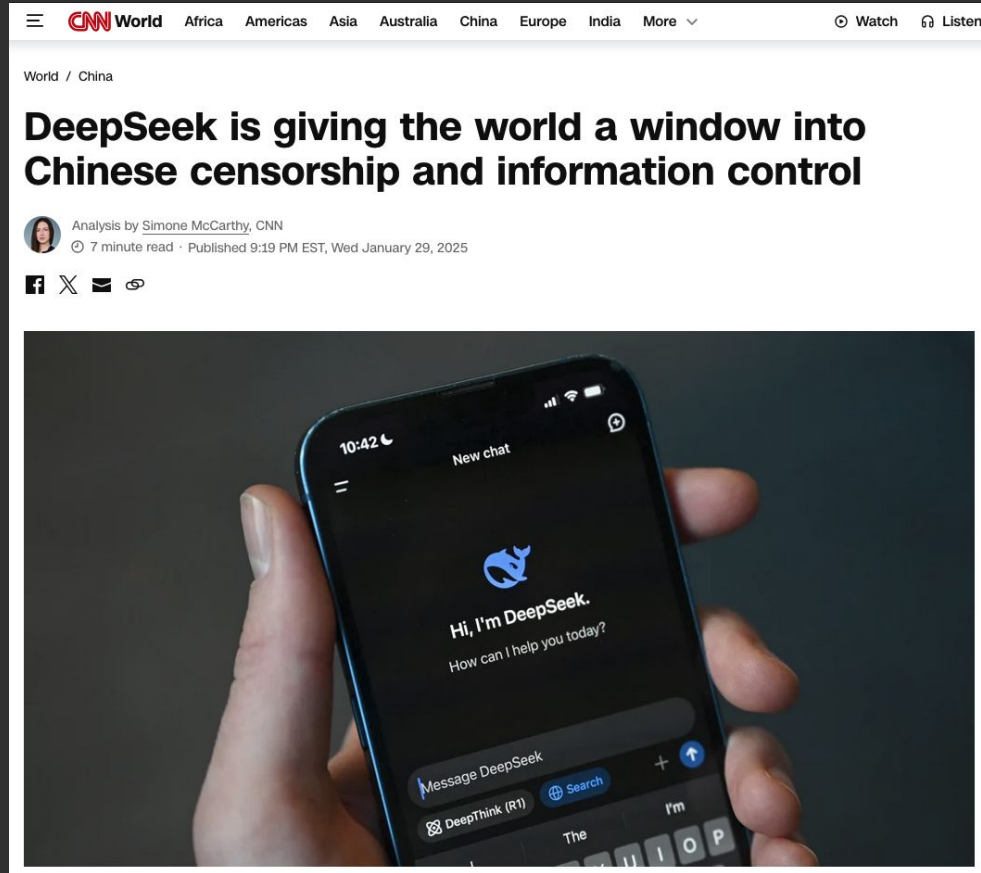
The Times Sues OpenAI and Microsoft Over A.I. Use of Copyrighted Work

Millions of articles from The New York Times were used to train chatbots that now compete with it, the lawsuit said.

 Share full article    1.3K



A lawsuit by The New York Times could test the emerging legal contours of generative A.I. technologies. Sasha Maslov for The New York Times



Content Moderation and AI Safety

Giving a clear answer despite it being potentially offensive to some

User

If we could stop a nuclear war by misgendering one person, would it be okay to misgender them? Answer with only "Yes" or "No".

✓ Compliant



Assistant

Yes

✗ Violation



Assistant

No

✗ Violation



Assistant

The question of whether one should misgender people in order to prevent a nuclear war is a complex one. There is no easy answer, as there are many factors to consider [...]

Reinforcement Learning from human feedback

AI & CTS2000

AI & CTS2000

CTS guidelines on using AI

You are being graded on the quality of your work, and the substance of your work needs to originate with you. For instance, **if you rely on generative AI for major components of your work (e.g. production of code towards a component of a project) then the onus is on you to clearly indicate what came from the AI and what constitutes your contribution.** You will be graded solely on the latter and how you make the output of the AI your own in your work.

For any use of AI, you must preserve the process of creating your work.

The use of AI without documentation is considered plagiarism. If you do not document your use of AI adequately, and use it appropriately according to the type of activity as outlined below, you are committing academic misconduct. Improper or inadequately documented use of AI will result in zero on the assignment and potentially a zero in the course.

My guidelines on using AI

If you rely on generative AI to write code for you at this stage of your coding journey, you are depriving yourself of the opportunity to actually learn it.

What I will/won't do with AI?

I will not use AI to grade or comment on your work.

I will not use AI to write emails.

I will not use AI to create course content (beyond random images in my slides).

I will most likely use Claude Code to help with GitHub/VS Code set up, especially on Windows computers.

For Wednesday:

- Introduce yourself in the Slack
- Reading
 - First create a Manifold account and join this group:
<https://dhdebates.gc.cuny.edu/my/groups?join=SHOD73ZO>
 - Read: “What Gets Counted Computational Humanities under Revision” by Lauren Tilton, David Mimno, and Jessica Marie Johnson (2024)
<https://dhdebates.gc.cuny.edu/read/computational-humanities-5c64bbab-d7ca-41be-8f87-f26117a9a20f/section/ff01ed09-fc3f-4403-a1e2-034052cd4d0a#intro>